

MC8P MIDI Controller - Owner's Manual

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Contents

- Introduction
- User Guide
 - 1.0 Interface Navigation
 - 3.0 Settings
 - 3.1 Assign Settings
 - 3.2 Display Settings
 - 3.3 About page
 - 3.4 Factory reset
- A Note of Thanks

Introduction

The MC8P MIDI Controller is a compact, MIDI device designed to enhance hands-on control of the OP-XY by *teenage engineering*. Featuring 8 fully assignable potentiometers, this controller provides immediate access to all MIDI assigned parameters—reducing menu-diving and streamlining your live performance workflow.

Built around the Teensy 4.0 microcontroller, the MC8P is an open-source project, offering both reliability and customization. I developed this controller out of necessity, seeking a more fluid and intuitive way to manipulate the OP-XY in real time. Whether tweaking filters, modulating effects, or shaping envelopes, the MC8P puts essential controls at your fingertips.

This manual will guide you through setup, configuration, and operation—helping you unlock the full potential of your MC8P.

User Guide

1.0 Interface Navigation

The MC8P controller features four buttons for navigation and eight potentiometers for control. The 128x64 OLED display provides visual feedback for all operations.

Buttons:

- **ASSIGN** - Enter MENU screen, navigate backwards, cancel actions, return to previous screens
- **ENTER** - Confirm selections, activate TEMP mode in the PERFORMANCE mode
- **PREV** - Move up in menu screens or decrease values
- **NEXT** - Move down in menu screens or increase values

1.1 Screen Hierarchy

The interface is organized into several screens accessible from the main display:
MAIN SCREEN → MENU → SETTINGS or CC ASSIGN → various configuration screens

1.1.1 Main Screen - PERFORMANCE Mode

This is your default operating view. The screen displays eight circular indicators, one for each potentiometer, showing current MIDI values (0-127). Each circle fills proportionally to the value being sent.

From the main screen:

- Press the **ASSIGN** button to open the menu
- Press and hold the **ENTER** button to activate temporary override mode (the display inverts as visual feedback)

TEMP Mode

When you press and hold ENTER on the main screen, the controller enters temporary override mode. In this mode, all potentiometer adjustments are temporary. When you release ENTER, the controller returns to the stored values and activates catch up mode, where potentiometers must pass through their stored positions before taking control again.

Use case example:

You can temporarily swell filter cutoffs, resonance, and effects sends to create a rising build-up, then release ENTER to snap everything back to your original mix - all without ever losing your starting point.

Note:

TEMP Mode does **not** save any changes to your controller's memory. It is purely for live, non-destructive parameter manipulation.

1.1.2 ASSIGN Screen - CC Assign

This screen lets you configure up to 10 MIDI messages per potentiometer.

Navigation flow:

1. Select which potentiometer to edit (flashing number)
2. Select which message slot to edit, or choose Add/Remove
3. Edit individual parameters

Parameters you can edit for each message:

- Channel – MIDI channel (1-16) or OMNI
- CC – Control Change number (0-127)
- Min Value – Minimum output value (0-127)
- Max Value – Maximum output value (0-127)
- Direction – Normal (0-127) or Inverted (127-0)

Editing values: When editing a parameter, use potentiometer 8 to smoothly adjust the value across its full range. The selected parameter flashes to indicate edit mode.

Navigation within Assign Screen:

- PREV/NEXT – Navigate between pots, messages, or Add/Remove buttons
- ENTER – Enter edit mode for a parameter, or cycle to the next parameter
- ASSIGN – Return to pot selection, or save and exit (confirmation popup appears)

Add/Remove messages:

- Navigate to "+ Add" to create a new message (up to 10 per pot)
- Navigate to "- Remove" to delete the selected message (minimum 1 message per pot)

1.1.3 Settings Screen

From the menu, select Settings to access:

- Assign Settings – Choose which potentiometer (1-8) to use for editing values in the Assign screen. Use PREV/NEXT to change the selection.
- Display Settings – Toggle the display inversion between normal and inverted. Press ENTER to enter edit mode, then use PREV or NEXT to toggle between Yes and No. Press ENTER again to save, or ASSIGN to cancel.
- About – View firmware version and device information.
- Factory Reset – Restore all settings to default values.

A Note of Thanks

Thank you for purchasing the MC8P MIDI Controller and supporting this passion project of mine! As someone who loves making music and tinkering with hardware, creating this controller has been a deeply rewarding experience and I'm incredibly grateful that you've chosen to be part of its journey.

The MC8P was born out of my own need for a more tactile, performance-friendly way to control the OP-XY, and I hope it brings the same fluidity and joy to your setup as it has to mine.

Your Feedback Matters

This is an evolving project, and I'd love to hear from you! If you have ideas for new features, improvements, or just want to share how you're using the MC8P, please reach out via:

Email: axs.instruments@gmail.com

Instagram: [@axs.instruments](https://www.instagram.com/axs.instruments)

GitHub: [MC8P Midi Controller](https://github.com/axs.instruments/MC8P-Midi-Controller)

Your input helps shape future updates, so don't hesitate to get in touch.

Most of all... have fun! I hope the MC8P becomes an indispensable part of your creative process.